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# Thermodynamic assessment of Ag–Dy–Sb ternary system

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ABSTRACT

Thermodynamic description of the Ag–Dy–Sb ternary system has been performed. At first, the boundary binary Dy–Sb system has been thermodynamically assessed with CALPHAD method. The solution phases including liquid, Bcc and Hcp were described with a substitutional model, of which the excess Gibbs energies were formulated with the Redlich–Kister polynomial. All the binary intermetallic phases, i.e. $Dy_5Sb_3$, $Dy_4Sb_3$ and DySb, were treated as stoichiometric compounds. Then, combined with the thermodynamic parameters of the Ag–Dy and Ag–Sb binary systems cited from literatures, the Ag–Dy–Sb ternary system was thermodynamically assessed. The calculated phase equilibria were in good agreement with the reported experimental data.

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## 1. Introduction

Silver with high electronic-conductivity has found wide application in electronic industry. However, pure Ag is of poor strength and cannot withstand high temperature. Fortunately, addition of antimony and rare earth elements into Ag can enhance its strength [1–2]. Besides, the melting points of many Ag-RE intermetallic compounds are very high. These compounds can improve the stability of Ag-base alloy at high temperature [2]. In order to provide useful information for develop new materials to satisfy new usages, phase diagrams of the Ag-related system are necessary. In this work, the Ag–Dy–Sb ternary system is assessed through CALPHAD approach.

## 2. Experiment information

### 2.1. The Dy–Sb binary system

Mironov et al. [3] have constructed phase diagram of the Dy–Sb system by using thermal analysis, X-ray diffraction and microstructure observation. Three compounds, $Dy_5Sb_3$, $Dy_4Sb_3$ and DySb, have been detected. Among these compounds, $Dy_5Sb_3$ and $Dy_4Sb_3$ form through peritectic reactions at 1953 K and 2053 K, respectively, whereas DySb melts congruently at 2443 K.

According to Mironov et al. [3], $Dy_4Sb_3$ could be stable from 2053 K down to room temperature. However, this result may be dubious according to the later studies [2,4–7]. On the basis of the experimental investigations [2,4–7], $Dy_4Sb_3$ decomposes eutectoidly into $Dy_5Sb_3$ and DySb at low temperature. Especially, Morozkin's experimental result [4] indicated that $Dy_4Sb_3$ was unstable at 1100 K. So, the eutectoid temperature is above 1100 K. Experimental information in Refs. [2,4–7] may be more reliable and is given higher weight in the present assessment.

Besides, Mironov et al. [3] indicated that the liquidus in the region of compositions with more than 50 at.% Sb was difficult to be measured reliably because of the high volatility of Sb at high temperature. So, lower weight has been given to the liquidus in the region of compositions with more than 50 at.% Sb during assessment.

As for thermodynamic properties, only the formation enthalpies of the three compounds, i.e. $Dy_5Sb_3$, $Dy_4Sb_3$ and DySb, were reported. By calorimetric measurement along with X-ray diffraction and metallographic analyses, Ferro et al. [8] measured the formation enthalpies of $Dy_5Sb_3$, $Dy_4Sb_3$ and DySb at 300 K. These thermodynamic data will be adopted in the present optimization.

### 2.2. The Ag–Dy and Ag–Sb binary systems

The Ag–Dy binary system has recently been assessed by Long et al. (unpublished work). The assessed phase diagram of Ag–Dy was shown in Fig. 1. Their thermodynamic parameters can reproduce most experimental phase diagram and thermodynamic properties well. Hence, the parameters assessed by Long et al. were directly adopted in present work.

Thermodynamic assessment of the Ag–Sb system had been carried out by Lee et al. [9]. However, their parameters could not well reproduce the experimental phase diagram in the temperature range below 500 K, and the calculated heats of formation of the solid phases showed large discrepancies from the experimental ones. Later, Oh et al. [10] reassessed this binary system, and most experimental data including the phase diagram at low temperature

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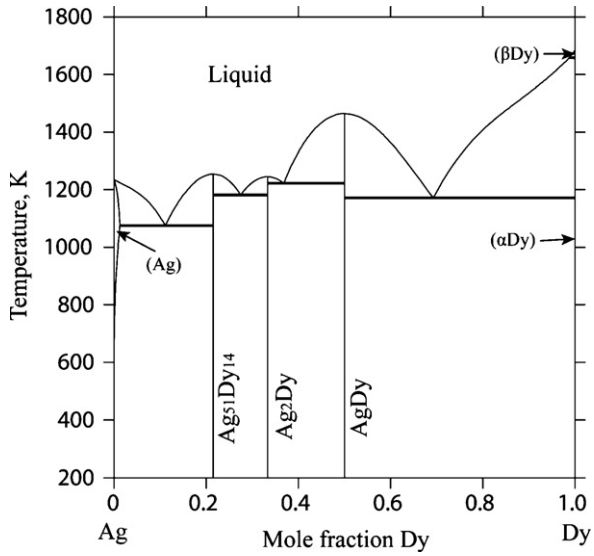


**Fig. 1.** The calculated Ag–Dy phase diagram by Long et al.

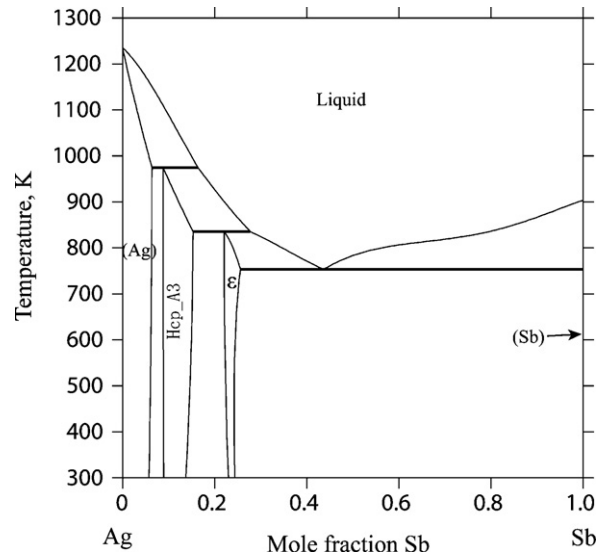


**Fig. 2.** The calculated Ag–Sb phase diagram [10].

**Table 1**
Optimized parameters of Ag–Dy–Sb ternary system.

| Phase | Thermodynamic parameters | References |
|---|---|---|
| Liquid | ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ag,Dy}} = -120,882.06 + 21.174T$ | Long et al. |
| | ${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ag,Dy}} = -69,962.11 + 27.008T$ | Long et al. |
| | ${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ag,Dy}} = 27{,}773.12$ | Long et al. |
| | ${}^{3}L^{\mathrm{Liq}}_{\mathrm{Ag,Dy}} = 18,181.39$ | Long et al. |
| | ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ag,Sb}} = -821.8 - 9.656T$ | [10] |
| | ${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ag,Sb}} = -19,309 + 4.424T$ | [10] |
| | ${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ag,Sb}} = -10,381.2$ | [10] |
| | ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Dy,Sb}} = -272,808.51 - 24.975T$ | This work |
| | ${}^{1}L^{\mathrm{Liq}}_{\mathrm{Dy,Sb}} = 16,379.60 - 12.158T$ | This work |
| | ${}^{2}L^{\mathrm{Liq}}_{\mathrm{Dy,Sb}} = 88,463.43$ | This work |
| Fcc | ${}^{0}L^{\mathrm{Fcc}}_{\mathrm{Ag,Dy}} = -65,136.13 - 5.743T$ | Long et al. |
| | ${}^{0}L^{\mathrm{Fcc}}_{\mathrm{Ag,Sb}} = -22,937.9 + 54.185T$ | [10] |
| | ${}^{1}L^{\mathrm{Fcc}}_{\mathrm{Ag,Sb}} = -530.7 - 58.387T$ | [10] |
| Bcc | ${}^{0}L^{\mathrm{Bcc}}_{\mathrm{Dy,Sb}} = -210,000$ | This work |
| Hcp | ${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Ag,Dy}} = 100,000$ | Long et al. |
| | ${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Ag,Sb}} = -23,050.3 + 38.368T$ | [10] |
| | ${}^{1}L^{\mathrm{Hcp}}_{\mathrm{Ag,Sb}} = -5357.5 - 45.064T$ | [10] |
| | ${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Dy,Sb}} = -200,000$ | This work |
| $Ag_{51}Dy_{14}$ | ${}^{0}G^{\mathrm{Ag_{51}Dy_{14}}}_{\mathrm{Ag:Dy}} = 0.7846{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Ag}} + 0.2154{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} - 23939.75 + 1.445T$ | Long et al. |
| $Ag_2Dy$ | ${}^{0}G^{\mathrm{Ag_2Dy}}_{\mathrm{Ag:Dy}} = 0.6667{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Ag}} + 0.3333{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} - 31050 + 2.207T$ | Long et al. |
| AgDy | ${}^{0}G^{\mathrm{AgDy}}_{\mathrm{Ag:Dy}} = 0.5{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Ag}} + 0.5{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} - 35,003.2 + 2.704T$ | Long et al. |
| $\varepsilon$ | ${}^{0}G^{\mathrm{Ortho}}_{\mathrm{Ag:Ag}} = {}^{0}G^{\mathrm{Fcc}}_{\mathrm{Ag}} + 4750 - 0.5T$ | [10] |
| | ${}^{0}G^{\mathrm{Ortho}}_{\mathrm{Ag:Sb}} = 0.75{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Ag}} + 0.25{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} - 338.5 - 4.5184T$ | [10] |
| | ${}^{0}G^{\mathrm{Ortho}}_{\mathrm{Sb:Ag}} = 0.25{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Ag}} + 0.75{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} + 23,223.9$ | [10] |
| | ${}^{0}G^{\mathrm{Ortho}}_{\mathrm{Sb:Sb}} = {}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} + 20,000 - 10T$ | [10] |
| | ${}^{0}L^{\mathrm{Ortho}}_{\mathrm{Ag:(Ag,Sb)}} = -4873.3$ | [10] |
| | ${}^{0}L^{\mathrm{Ortho}}_{\mathrm{(Ag,Sb):Sb}} = +5932.3$ | [10] |
| $Dy_5Sb_3$ | ${}^{0}G^{\mathrm{Dy_5Sb_3}}_{\mathrm{Dy:Sb}} = 0.625{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} + 0.375{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} - 99,569.55 + 1.266T$ | This work |
| $\beta$_$Dy_4Sb_3$ | ${}^{0}G^{\mathrm{Dy_4Sb_3_H}}_{\mathrm{Dy:Sb}} = 0.5714{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} + 0.4286{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} - 106,742.07 + 0.039T$ | This work |
| $\alpha$_$Dy_4Sb_3$ | ${}^{0}G^{\mathrm{Dy_4Sb_3_L}}_{\mathrm{Dy:Sb}} = 0.5714{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} + 0.4286{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} - 107,715 + 0.55T$ | This work |
| $\beta$_DySb | ${}^{0}G^{\mathrm{DySb_H}}_{\mathrm{Dy:Sb}} = 0.5{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} + 0.5{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} - 117,655.46 + 0.035T$ | This work |
| $\alpha$_DySb | ${}^{0}G^{\mathrm{DySb_L}}_{\mathrm{Dy:Sb}} = 0.5{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} + 0.5{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} - 120,496.2 + 1.3486T$ | This work |
| $\tau_1$ | ${}^{0}G^{\tau_1}_{\mathrm{Ag:Dy:Sb}} = 0.38462{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Ag}} + 0.53846{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} + 0.07692{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} - 48,000$ | This work |
| $\tau_2$ | ${}^{0}G^{\tau_2}_{\mathrm{Ag:Dy:Sb}} = 0.25{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Ag}} + 0.25{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Dy}} + 0.5{}^{0}G^{\mathrm{Rho}}_{\mathrm{Sb}} - 62,250$ | This work |

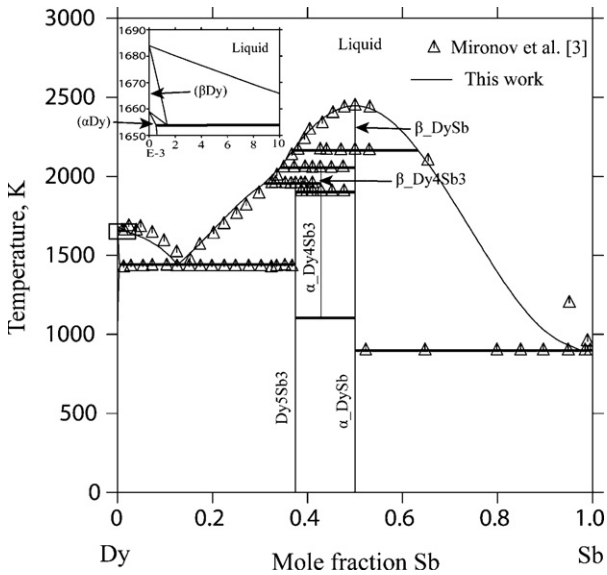


**Fig. 3.** The calculated Dy–Sb phase diagram with the experimental data [3].

and heats of formation of the solid phases were well reproduced. So, the thermodynamic parameters reported by Oh et al. [10] were adopted in this work. Fig. 2 illustrates the calculated phase diagram based on Oh et al.'s assessment.

### 2.3. The Ag–Dy–Sb ternary system

Zeng and Xie [7] has established the isothermal section of this ternary system at 673 K by powder X-ray diffraction with the aid of optical microscopy and scanning electron microscopy. Two ternary compounds, i.e. $Ag_5Dy_7Sb$ and $AgDySb_2$, were detected and their crystal structures determined. No remarkable solubility of the third element in the binary phases was detected.

## 3. Thermodynamic model

### 3.1. Solution phases

A substitutional solution model based on random mixing of the constituent atoms is employed to describe liquid, Bcc, Fcc and Hcp solutions. The molar Gibbs energy of a solution phase $\Phi$ ($\Phi$ = liquid, Bcc, Fcc, Hcp) is expressed as:

$$G^{\Phi} = \sum_{i=\mathrm{Ag,Dy,Sb}} x_i^0 G_i^{\Phi} + RT \sum_{i=\mathrm{Ag,Dy,Sb}} x_i \ln x_i + {}^{E}G_m^{\Phi} \quad (1)$$

where

$$\begin{aligned} {}^{E}G_m^{\Phi} = {} & x_{\mathrm{Ag}}x_{\mathrm{Dy}} \sum_{j=0,1,\ldots}^{N} (x_{\mathrm{Ag}} - x_{\mathrm{Dy}})^{j\,(j)}L_{\mathrm{Ag,Dy}}^{\Phi} + x_{\mathrm{Dy}}x_{\mathrm{Sb}} \\ & \sum_{j=0,1,\ldots}^{N} (x_{\mathrm{Dy}} - x_{\mathrm{Sb}})^{j\,(j)}L_{\mathrm{Dy,Sb}}^{\Phi} + x_{\mathrm{Ag}}x_{\mathrm{Sb}} \\ & \sum_{j=0,1,\ldots}^{N} (x_{\mathrm{Ag}} - x_{\mathrm{Sb}})^{j\,(j)}L_{\mathrm{Ag,Sb}}^{\Phi} + x_{\mathrm{Ag}}x_{\mathrm{Dy}}x_{\mathrm{Sb}}L_{\mathrm{Ag,Dy,Sb}}^{\Phi} \end{aligned} \quad (2)$$

here $\Phi$ denotes the solution phases, $x_i$ ($i$ = Ag, Dy and Sb) means mole fraction of component $i$, and ${}^{0}G_i^{\Phi}$ is the molar Gibbs energy of pure element $i$ in the structural state of $\Phi$. ${}^{(j)}L_{\mathrm{Ag,Dy}}^{\Phi}$ is taken from Long et al. and ${}^{(j)}L_{\mathrm{Ag,Sb}}^{\Phi}$ from Oh et al. [10]. ${}^{(j)}L_{\mathrm{Dy,Sb}}^{\Phi}$ is expressed as:

$${}^{(j)}L_{\mathrm{Dy,Sb}}^{\Phi} = A_j + B_j T \quad (3)$$

where $A_j$ and $B_j$ are constants to be optimized. $L_{\mathrm{Ag,Dy,Sb}}^{\Phi}$ is the ternary interaction parameter and set to zero due to lack of experimental data.

**Table 2**
Comparison of invariant reactions in the Dy–Sb system.

| Reaction | Composition of liquid ($X_{Sb}$) | $T$ (K) | Type | References |
|---|---|---|---|---|
| L ↔ Hcp(Dy) + $Dy_5Sb_3$ | 0.126 | 1441 | Eutectic | This work |
| | 0.145 | 1433 | Eutectic | [3] |
| L ↔ α_DySb + Rho | 0.975 | 898 | Eutectic | This work |
| | 0.99 | 895 | Eutectic | [3] |
| L + Bcc(Dy) ↔ Hcp(Dy) | 0.018 | 1654 | Peritectic | This work |
| | 0.075 | 1653 | Peritectic | [3] |
| L + β_$Dy_4Sb_3$ ↔ $Dy_5Sb_3$ | 0.309 | 1953 | Peritectic | This work |
| | 0.325 | 1953 | Peritectic | [3] |
| L + α_DySb ↔ β_$Dy_4Sb_3$ | 0.342 | 2053 | Peritectic | This work |
| | 0.35 | 2053 | Peritectic | [3] |
| L + β_DySb ↔ α_DySb | 0.369 | 2163 | Peritectic | This work |
| | 0.38 | 2163 | Peritectic | [3] |
| L + β_DySb ↔ α_DySb | 0.632 | 2163 | Peritectic | This work |
| | 0.63 | 2163 | Peritectic | [3] |
| L ↔ β_DySb | 0.5 | 2445 | Congruent | This work |
| | 0.5 | 2443 | Congruent | [3] |
| β_$Dy_4Sb_3$ ↔ α_$Dy_4Sb_3$ | – | 1900 | Allotropic | This work |
| | – | 1903 | Allotropic | [3] |
| β_DySb ↔ α_DySb | – | 2163 | Allotropic | This work |
| | – | 2163 | Allotropic | [3] |
| α_$Dy_4Sb_3$ ↔ $Dy_5Sb_3$ + α_DySb | – | 1103 | Eutectoidal | This work |
| | – | >1100 | Eutectoidal | [4] |

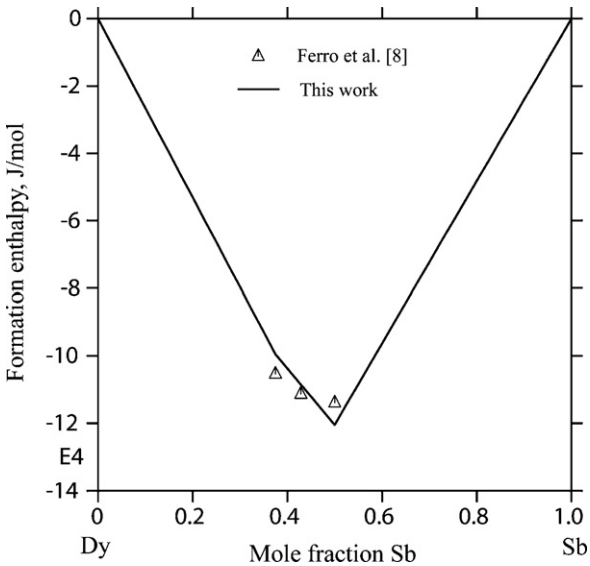


**Fig. 4.** Calculated formation enthalpies of the compounds compared with experimental data [8]. Reference states: Hcp_A3 Dy and Rhombohedral_A7 Sb.

### 3.2. Binary intermetallic compounds

All the binary intermetallic phases except $\varepsilon$ in the Ag–Sb system are treated as stoichiometric phases, which are described as $M_pN_q$. Because the experimental data on heat capacity is unavailable, the Gibbs energy of the phase $M_pN_q$ is simply formulated by using Neumann–Kopp rule as:

$$G_{M_pN_q} = \frac{p}{p+q}{}^0G_M^{\mathrm{HSER}} + \frac{q}{p+q}{}^0G_N^{\mathrm{HSER}} + A + BT \quad (4)$$

where $M$ and $N$ denote Ag, Dy or Sb, $A$ and $B$ are the adjusted parameters.

The intermediate phase $\varepsilon$ in the Ag–Sb system has an orthorhombic structure with a β-$Cu_3Ti$ prototype, which exhibits a rather wide range of homogeneity. Gibbs energy of this phase is directly taken from Oh et al. [10].

### 3.3. Ternary intermetallic compounds

There is no report about the homogeneity range of the ternary phases. So, $\tau_1$-$Ag_5Dy_7Sb$ and $\tau_2$-$AgDySb_2$ are modeled as stoichiometric phases, $Ag_xDy_ySb_z$, of which the Gibbs energy are modeled as following:

$$G_{Ag_xDy_ySb_z} = \frac{x}{x+y+z}{}^0G_{Ag}^{\mathrm{Fcc}} + \frac{y}{x+y+z}{}^0G_{Dy}^{\mathrm{Hcp}} + \frac{z}{x+y+z}{}^0G_{Sb}^{\mathrm{Rho}} + C + DT \quad (5)$$

here $C$ and $D$ are the parameters to be optimized in this work.

## 4. Results and discussion

### 4.1. The Dy–Sb binary system

With the lattice stabilities cited from Dinsdale [11], thermodynamic assessment of the Dy–Sb system was carried out by using the Thermo_Calc program and the parameters were listed in Table 1, which reproduce most experimental data well as can be seen in the next paragraphs.

Fig. 3 shows the calculated phase diagram of the Dy–Sb system compared with experimental data [3], and the invariant reactions in the Dy–Sb system are listed in Table 2. It is easy to see that most experimental data for the diagram boundaries were well reproduced. The differences between the calculated and measured temperatures of all invariant equilibria are less than 8 K. Although at the Sb-rich side, the calculated liquidus curve slightly deviates from the experimental data, the assessed results in present work can still be accepted considering the experimental errors. As pointed out by Mironov et al. [3], due to the high volatility of Sb at high temperature, it is difficult to obtain reliable liquidus in the region with more than 50 at.% Sb. Because the composition of the sample may shift to Dy-rich side during heating compared to the normal one, the liquidus curves shift to Dy-rich side. Obviously, this deviation should decrease with decreasing the content of Sb.

Formation enthalpies of the compounds at 300 K were calculated in comparison with the experimental data as illustrated in Fig. 4. Apparently, the calculated values agree with the experimental data with reasonable deviations.

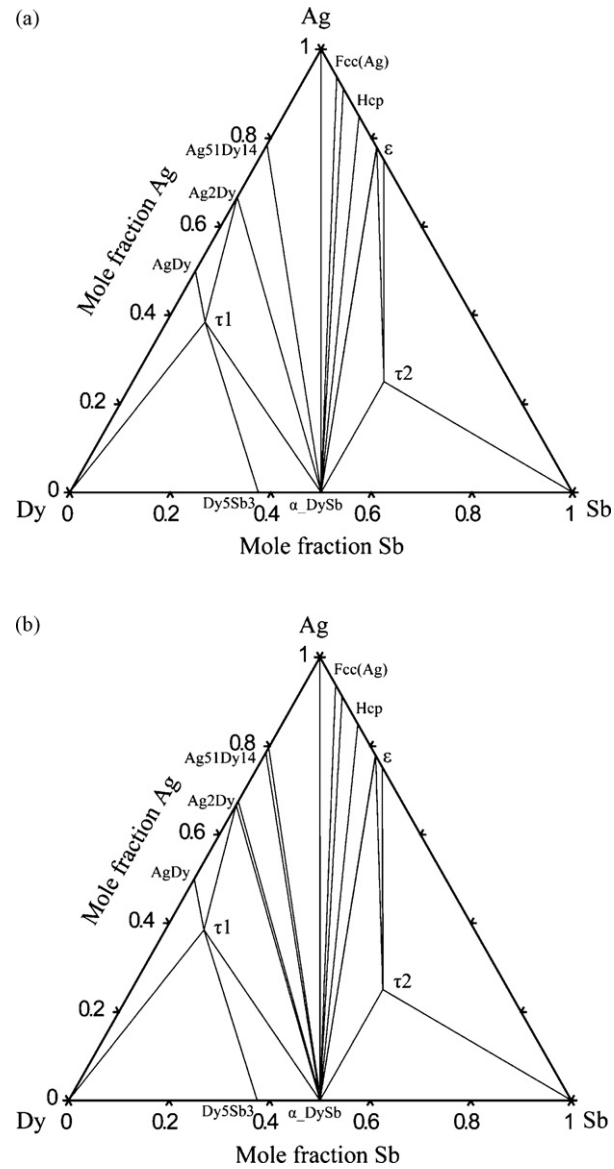


**Fig. 5.** The calculated (a) and experimental (b) [7] isothermal section of the Ag–Dy–Sb ternary system at 673 K.

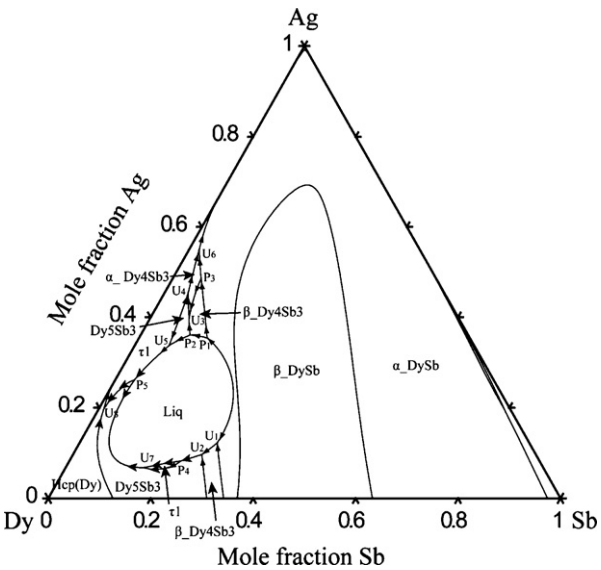


**Fig. 6.** The calculated liquidus projection of the Ag–Dy–Sb ternary system.

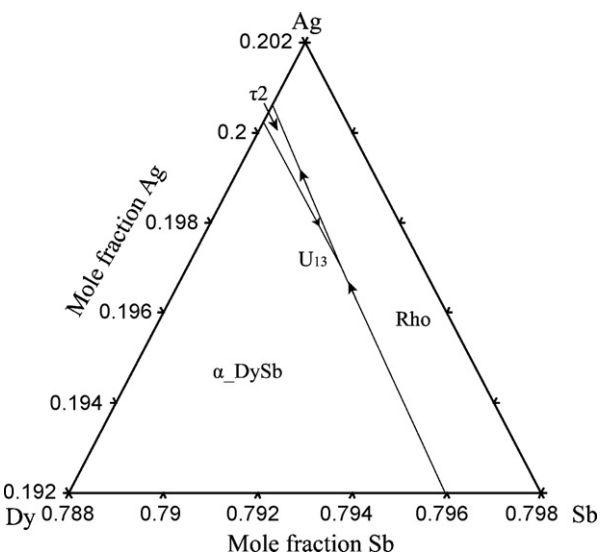


**Fig. 7.** Enlarged part of the liquidus projection in the Ag–Dy–Sb ternary system.

### 4.2. The Ag–Dy–Sb ternary system

By using the presently optimized parameters of the Dy–Sb system along with the reported parameters of the Ag–Dy and Ag–Sb [10] systems, and based on the ternary phase relations proposed by Zeng and Xie [7], phase equilibria in the Ag–Dy–Sb ternary system has been further extrapolated. The thermodynamic parameters are listed in Table 1. The calculated and experimental Ag–Dy–Sb isothermal sections at 673 K are shown in Fig. 5. It is easy to see that the stability of the compounds $Ag_5Dy_7Sb$ and $AgDySb_2$, and all the other experimental phase relations have been well reproduced.

Additionally, the liquidus projection is calculated as shown in Figs. 6 and 7. Invariant reactions $E_1$, $E_2$, $E_3$, $U_9$ and $U_{10}$ are very close to the Ag–Dy binary subsystem, and $E_4$, $U_{11}$, $U_{12}$, $U_{13}$ and $U_{14}$ are very near to the Ag–Sb binary subsystem, so, these 10 invariant reactions cannot be visible in Fig. 6. The calculated invariant reactions and temperatures involved liquid in the Ag–Dy–Sb ternary system are summarized in Table 3. Further experimental information is needed to verify the liquidus projection.

**Table 3**
Calculated invariant reactions and temperatures of the Ag–Dy–Sb ternary system.

| Type | Reaction | $T$ (K) |
|---|---|---|
| $E_1$ | $L \leftrightarrow \tau_1 + Ag_2Dy + AgDy$ | 1222.5 |
| $E_2$ | $L \leftrightarrow \alpha_DySb + Ag_2Dy + Ag_{51}Dy_{14}$ | 1182.1 |
| $E_3$ | $L \leftrightarrow \alpha_DySb + Ag_{51}Dy_{14} + Fcc(Ag)$ | 1074.8 |
| $E_4$ | $L \leftrightarrow \tau_2 + Rho + \varepsilon$ | 753.2 |
| $U_1$ | $L + \alpha_DySb \leftrightarrow \beta_Dy_4Sb_3 + L$ | 2002.8 |
| $U_2$ | $L + \beta_Dy_4Sb_3 \leftrightarrow L + Dy_5Sb_3$ | 1917.8 |
| $U_3$ | $L + \beta_Dy_4Sb_3 \leftrightarrow Dy_5Sb_3 + \alpha_Dy_4Sb_3$ | 1903 |
| $U_4$ | $L + Dy_5Sb_3 \leftrightarrow \tau_1 + \alpha_Dy_4Sb_3$ | 1851 |
| $U_5$ | $L + Dy_5Sb_3 \leftrightarrow \tau_1 + L$ | 1830.6 |
| $U_6$ | $L + \alpha_Dy_4Sb_3 \leftrightarrow \tau_1 + \alpha_DySb$ | 1752.4 |
| $U_7$ | $L + \tau_1 \leftrightarrow L + Dy_5Sb_3$ | 1630.1 |
| $U_8$ | $L + Dy_5Sb_3 \leftrightarrow \tau_1 + Hcp(Dy)$ | 1372.4 |
| $U_9$ | $L + \alpha_DySb \leftrightarrow \tau_1 + Ag_2Dy$ | 1245 |
| $U_{10}$ | $L + \tau_1 \leftrightarrow AgDy + Hcp(Dy)$ | 1171.2 |
| $U_{11}$ | $L + Fcc(Ag) \leftrightarrow \alpha_DySb + Hcp(Dy)$ | 974.4 |
| $U_{12}$ | $L + Hcp(Dy) \leftrightarrow \alpha_DySb + \varepsilon$ | 835.5 |
| $U_{13}$ | $L + \alpha_DySb \leftrightarrow \tau_2 + Rho$ | 831 |
| $U_{14}$ | $L + \alpha_DySb \leftrightarrow \tau_2 + \varepsilon$ | 818.1 |
| $P_1$ | $L + \alpha_DySb \leftrightarrow \beta_Dy_4Sb_3$ | 2002.8 |
| $P_2$ | $L + \beta_Dy_4Sb_3 \leftrightarrow Dy_5Sb_3$ | 1917.8 |
| $P_3$ | $L + \beta_Dy_4Sb_3 + \alpha_DySb \leftrightarrow \alpha_Dy_4Sb_3$ | 1903 |
| $P_4$ | $L + Dy_5Sb_3 \leftrightarrow \tau_1$ | 1830.6 |
| $P_5$ | $L + \tau_1 \leftrightarrow Dy_5Sb_3$ | 1630.1 |

## 5. Conclusion

With the CALPHAD method, the Dy–Sb binary system was optimized, and thermodynamic description of the Ag–Dy–Sb ternary system has been performed. Good agreement between the calculated and experimental data has been realized and a set of thermodynamic parameters for describing the Ag–Dy–Sb ternary system have been obtained.

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